CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

PROPOSAL PRESENTATION

# Planner Copilot

## A local AI assistant for production planning simulations

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Team 1 · Guided by Dr. Hamidreza Heidari · 20 October 2026

THE PROBLEM

# Good production planning needs simulation, and experts

A busy week, missing workers or a broken machine: a simulation shows what will happen before it does.

Today mostly only a simulation expert can run it correctly, and small companies often have none.

An AI chatbot alone cannot plan production, and nobody can check its answer without an expert.

## Key motivations

- Small companies get answers without a simulation expert
- Large companies save expert time
- Company data stays in-house
- Reliable AI help for planning is needed now

TECHNISCHE HOCHSCHULE DEGGENDORF

PROJECT SCOPE

# What we will and will not do

## Aim

Test whether a local AI assistant with code checks can answer planning questions at the level of published expert simulations.

## Includes

- About 4 published simulation cases
- Test questions with correct answers
- A local AI assistant with code checks
- Comparison of three setups

## Out of scope (current phase)

- A company’s own production data and models
- Commercial simulation tools
- Use in a real factory

GOALS

# Three measurable targets

## ≥ 90%

### Close to the expert

For questions with an answer, the AI’s answer is close to the expert’s answer.

## ≤ 2%

### Confidently wrong

The AI gives a wrong answer and sounds sure. This is the most dangerous error.

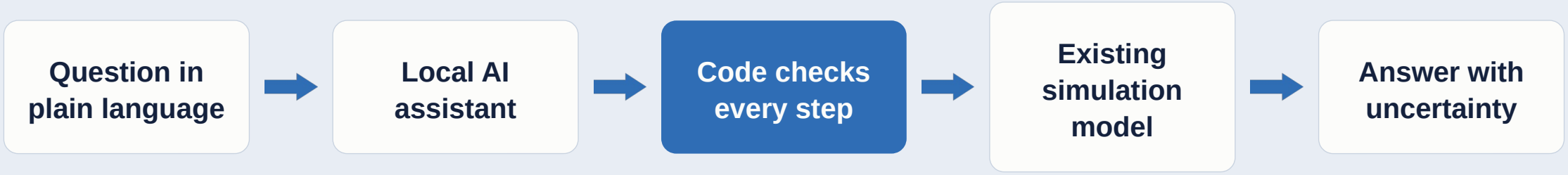
## ≥ 90%

### Correct refusals

When a question cannot be answered, the AI says so instead of guessing.

OUR IDEA

# An AI assistant that runs the simulation, and checks itself




*Example: “Busy week, +20% orders. Do we need a fourth worker?”*

## TECHNOLOGY STACK

| Process | Tool | Purpose |
|---|---|---|
| Simulation | JaamSim, Python | A computer model of a factory: shows what happens if orders, workers or machines change |
| AI model | Open-source LLM (Ollama) | The AI that understands plain-language questions; runs on our own computer |
| Code checks | Python, MCP server | Checks every AI step and refuses unclear questions |
| Interface | Gradio | A simple web page to ask questions and see charts |
| Compute | Google Colab GPU | A cloud computer with a graphics card to run the AI |
| Code sharing | GitHub | A shared place to store and track our code |

Tools may change as the project progresses !

PLAN

# Four steps to get there

### 1 · Find cases

About 4 published simulation cases from different industries, using public data only.

### 2 · Rebuild and ask

Rebuild each case and write test questions with the correct answers.

### 3 · Build the assistant

A local AI assistant with code checks, built with open-source tools.

### 4 · Compare

Three setups: expert, AI + simulation, and AI + Planner Copilot + simulation.

TEAM

# Who does what

### Yu Han

Team lead

Case 2
Code checks (guardrail engine)

### Melisa Subasi

Questions & report

Case 1
Test questions
Final report editor

### Meron Hagos

Interface & literature

Case 3
User interface
Literature review

### Alejandro Rivas

Experiments

Case 4
Experiments
GitHub & demo

A case is one published factory simulation model with known expert results.

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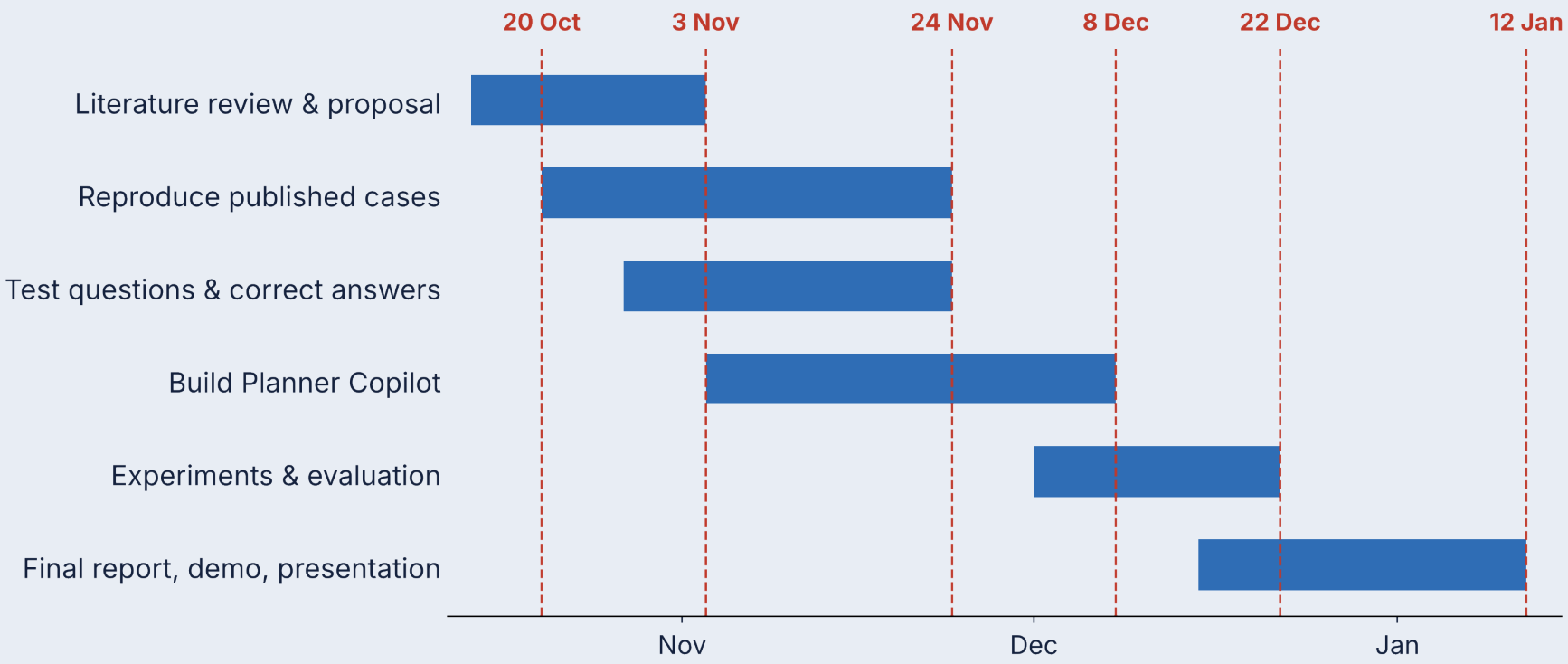
TIMELINE

# From proposal to final hand-in

| Date | Milestone |
|---|---|
| 20 Oct | Proposal, team, GitHub |
| 3 Nov | Literature report, method outline, baseline simulation |
| 24 Nov | Cases rebuilt, first version of the assistant |
| 8 Dec | First experiments |
| 22 Dec | Demo, experiments and analysis, report draft |
| 12 Jan | Final presentation, report, demo video |

GANTT CHART

# Project schedule




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EXPECTED OUTCOMES

# What we will hand in

### Test questions

Questions with correct answers for about 4 published cases.

### Planner Copilot

A working open-source assistant that runs the simulations.

### Comparison

Measured results for three setups against our targets.

### Report and demo

Final report, demo video and a GitHub repository.

BILL OF MATERIALS

# What the project needs

| Component | Specification | Cost |
|---|---|---|
| GPU access (Google Colab) | Cloud GPU for the local AI | Free tier, to be confirmed |
| Simulation cases and data | Published, public | Free |
| Software | Open source | Free |

RISKS, ETHICS

# Keeping it safe and realistic

## Ethics

Only public data, so no privacy issue. All sources are cited.

## Main risk

Rebuilding the cases may take longer than planned. Fallback: fewer cases.

# Thank You !

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github.com/bodhi584/planner-copilot